Digital Classroom Setup

Week 7 Assignment - Problem 4 (Intermediate)

Tablet inherits operate() from the abstract-class contract and implements Chargeable. The two charge methods demonstrate compile-time polymorphism by parameter count.

Java solution

```
abstract class ClassroomDevice {
    public abstract String operate();
}
interface Chargeable {
    String charge();
    String charge(int minutes);
}
class Tablet extends ClassroomDevice implements Chargeable {
    private final String assetTag;
    Tablet(String assetTag) { this.assetTag = assetTag; }
    @Override public String operate() {
        return "Tablet " + assetTag + " displaying lesson";
    }
    public String charge() {
        return assetTag + " charging";
    }
    public String charge(int minutes) {
        return assetTag + " charging for " + minutes + " minutes";
    }
}
public class Main {
    public static void main(String[] args) {
        Tablet t = new Tablet("TAB-5");
        System.out.println(t.operate());
        System.out.println(t.charge());
        System.out.println(t.charge(30));
    }
}
```

Sample output

```
Tablet TAB-5 displaying lesson
TAB-5 charging
TAB-5 charging for 30 minutes
```

Note: charge() and charge(int) are overloads; Java selects the method from the supplied argument list.